

1. Android Environment Setup

Create a simple Android application named *HelloMobile* using Android Studio. Configure the SDK and emulator, run the application on the emulator, and display “Hello Mobile Programming” on the screen.

Code:

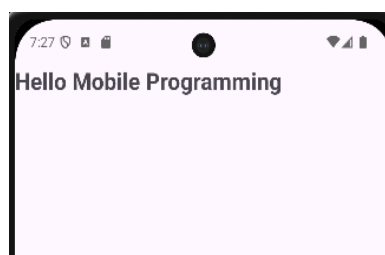
MainActivity.java

```
package com.example.hellomobile;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
.....
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        .....
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android=" "
    xmlns:app=" "
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello Mobile Programming"
        android:textSize="24sp"
        android:textStyle="bold"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output :



2. User Interface Design Using Layouts

Design an Android UI using LinearLayout, RelativeLayout (or ConstraintLayout), and include TextView, EditText, and Button. Set layout attributes such as orientation, margin, padding, and gravity.

Code:

MainActivity.java

```
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
.....
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        .....
        .....
    }
}
```

activity_main.xml(ConstraintLayout)

```
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android=" " xmlns:app=" "
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello Mobile Programming"
        android:textSize="24sp"
        android:layout_margin="16dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

    <EditText
        android:id="@+id/etName"
        android:layout_width="0dp"
        android:hint="Enter your name"
        android:layout_margin="16dp"
        app:layout_constraintTop_toBottomOf="@id/tvTitle"
        ...../>

    <Button
        android:text="Submit"
```

```

        android:layout_margin="16dp"
        app:layout_constraintTop_toBottomOf="@id/etName"
        ..... />
</androidx.constraintlayout.widget.ConstraintLayout>

```

activity_main.xml(LinearLayout)

```

<LinearLayout xmlns:android=" "
    android:layout_width="match_parent"
    android:layout_height="match_parent"
        android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello Mobile Programming"
        android:textSize="24sp"
        android:layout_margin="10dp" />

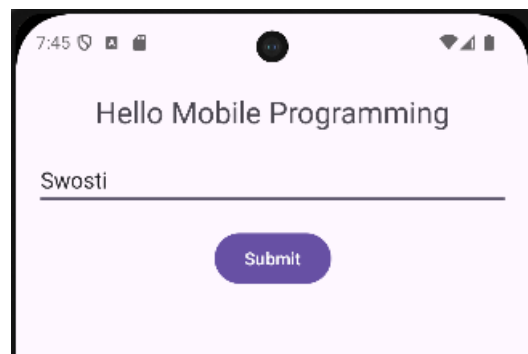
    <EditText
        android:hint="Enter your name"
        android:layout_margin="10dp"
        android:padding="10dp" />

    <Button
        android:text="Submit"
        android:layout_margin="10dp" />

</LinearLayout>

```

Output :



3. Event Handling and Resource Management

Create an Android application with a Button. When the button is clicked, display a message using string resources and change the background color using color resources. Use a drawable resource for the button icon.

Code:

Strings.xml

```
<resources>
    <string name="app_name">HelloMobile</string>
    <string name="button_text">Submit</string>
    <string name="message_text">Hello! Button clicked!</string>
</resources>
```

Colors.xml

```
<resources>
    <color name="white">#FFFFFF</color>
    <color name="light_blue">#87CEEB</color>
    <color name="button_color">#FF6200EE</color>
</resources>
```

ic_button_icon.xml

```
<vector xmlns:android=" "
.....>
    <path
        android:pathData="M12,2L15,8H9L12,2zM12,22a1,1 0,1 1,0-2 1,1 0,0 1,0 2z"/>
</vector>
```

MainActivity.java

```
import android.widget.*;
.....
.....
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        final LinearLayout mainLayout = findViewById(R.id.main);
        Button button = findViewById(R.id.buttonClick);
        button.setOnClickListener(new Button.OnClickListener() {
            @Override
            public void onClick(android.view.View v) {
                Toast.makeText(MainActivity.this, getString(R.string.message_text),
                Toast.LENGTH_SHORT).show();
                mainLayout.setBackgroundColor(ContextCompat.getColor(MainActivity.this,
                R.color.light_blue));
            }
        });
    }
}
```

Activity_main.xml

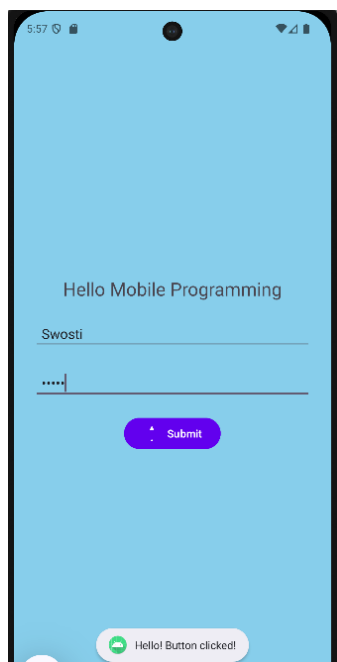
```
<LinearLayout xmlns:android=" "
```

```

        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/main"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="16dp"
        android:background="@color/white">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Mobile Programming"
            android:textSize="24sp"
            android:layout_margin="10dp" />
        <EditText
            android:layout_width="match_parent"
            android:hint="Enter your name"
            android:padding="10dp"
            ...../>
        <EditText
            android:hint="Enter your password"
            android:inputType="textPassword"
            ...../>
        <Button
            android:id="@+id/buttonClick"
            android:text="@string/button_text"
            android:drawableLeft="@drawable/ic_button_icon"
            android:backgroundTint="@color/button_color"
            android:textColor="@color/white"
            android:layout_margin="10dp" />
    </LinearLayout>

```

Output :



4. Activity and Intent Implementation

Develop an Android application with two activities. Enter text in the first activity and pass it to the second activity using an explicit intent. Display the received text in the second activity.

Code:

MainActivity.java

```
import android.widget.*;
.....
public class MainActivity extends AppCompatActivity {
    EditText editText;
    Button button;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        editText = findViewById(R.id.editTextMessage);
        button = findViewById(R.id.btnSend);
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String message = editText.getText().toString();
                Intent intent = new Intent(MainActivity.this, SecondActivity.class);
                intent.putExtra("MESSAGE_KEY", message);
                startActivity(intent);
            }
        });
    }
}
```

activity_main.xml

```
<LinearLayout xmlns:android=" "
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/editTextMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter text" />

    <Button
        android:id="@+id/btnSend"
        android:text="Send to Second Activity"
        android:layout_marginTop="10dp" />

</LinearLayout>
```

SecondActivity.java

```
import android.widget.*;
.....
public class SecondActivity extends AppCompatActivity {
    TextView textView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
        textView = findViewById(R.id.textViewResult);
        String receivedText = getIntent().getStringExtra("MESSAGE_KEY");
        textView.setText(receivedText);
    }
}
```

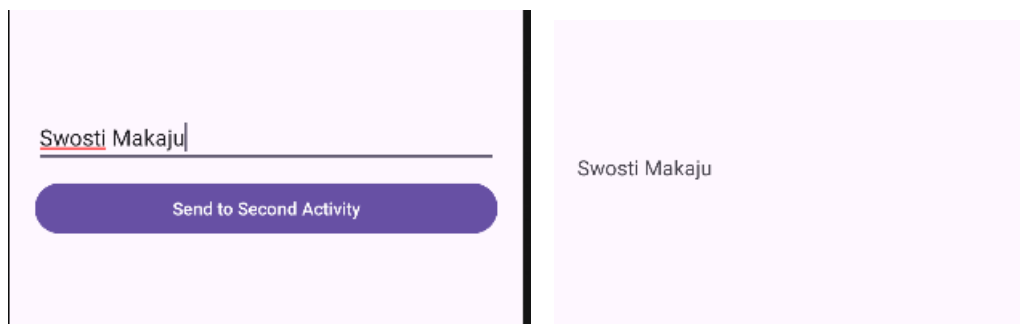
activity_second.xml

```
<LinearLayout xmlns:android=" "
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/textViewResult"
        android:layout_height="wrap_content"
        android:text="Received text appears in these activity"
        android:textSize="18sp"
        ..... />

</LinearLayout>
```

Output :



5. Fragment, Menu, and Dialog Implementation

Create an Android application using fragments. Implement an options menu and display an alert dialog when a menu item is selected.

Code:

MainActivity.java

```
import android.view.*;
.....
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        getSupportFragmentManager()
            .beginTransaction()
            .replace(R.id.fragment_container, new
com.example.hellomobile.MyFragment())
            .commit();
    }
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.menu_about) {
            showAlertDialog();
            return true;
        }

        if (id == R.id.menu_exit) {
            finish();
            return true;
        }

        return super.onOptionsItemSelected(item);
    }
    private void showAlertDialog() {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setTitle("About");
        builder.setMessage("This app demonstrates Fragment, Menu, and Alert Dialog
implementation.");
        builder.setPositiveButton("OK", null);
        builder.show();
    }
}
```


activity_main.xml

```
<FrameLayout
    android:id="@+id/fragment_container"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />
```

MyFragment.java

```
import androidx.fragment.app.Fragment;
import android.view.*;
.....
public class MyFragment extends Fragment {
    public MyFragment() { }
    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
        Bundle savedInstanceState) {
        return inflater.inflate(R.layout.fragment_my, container, false);
    }
}
```

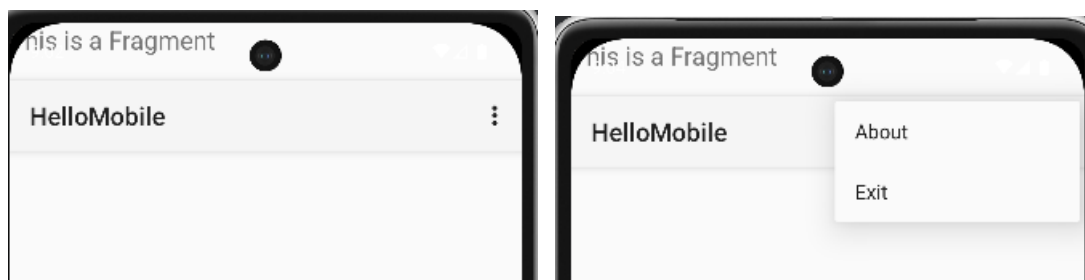
fragment_my.xml

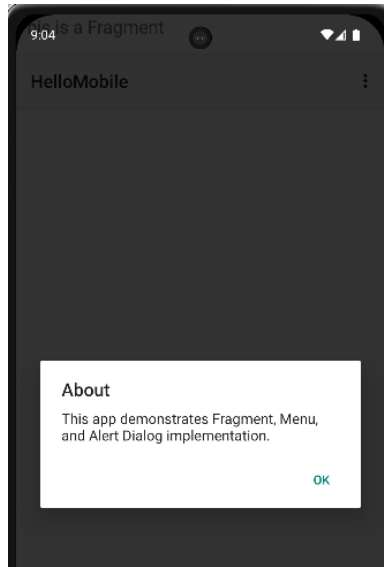
```
<FrameLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    .....>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is a Fragment"
        android:textSize="20sp"
        ..... /> </FrameLayout>
```

menu_main.xml

```
<menu >
    <item
        android:id="@+id/menu_about"
        android:title="About"
        app:showAsAction="never" />
    <item
        android:id="@+id/menu_exit"
        android:title="Exit"
        app:showAsAction="never" /> </menu>
```

Output :





6. RecyclerView Implementation

Create an Android application that displays a list of items (e.g., names or products) using RecyclerView with a custom adapter and ViewHolder.

Code:

MainActivity.java

```
import androidx.recyclerview.widget.*;
.....
public class MainActivity extends AppCompatActivity {
    RecyclerView recyclerView;
    String[] foodList = {"Apple", "Banana", "Cherry", "JackFruit",
"Mango","Orange"};
    String[] foodPriceList = {"500", "800", "400", "300","600", "100"};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        recyclerView = findViewById(R.id.recyclerView);
        recyclerView.setLayoutManager(new GridLayoutManager(this, 2));
        recyclerView.setAdapter(new MyAdapter(foodList, foodPriceList));
    }
}
```

activity_main.xml

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:fitsSystemWindows="true">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        ...../>
</LinearLayout>
```

MyAdapter.java

```
import android.view.*;
import android.widget.*;
.....
public class MyAdapter extends RecyclerView.Adapter<MyAdapter.ViewHolder> {
    String[] foodList;
    String[] foodPriceList;
    public MyAdapter(String[] foodList, String[] foodPriceList) {
        this.foodList = foodList;
        this.foodPriceList = foodPriceList;
    }
    public static class ViewHolder extends RecyclerView.ViewHolder {
        TextView txtName, txtPrice;
        public ViewHolder(View itemView) {
            super(itemView);
            txtName = itemView.findViewById(R.id.txtName);
        }
    }
}
```

```

        txtPrice = itemView.findViewById(R.id.txtPrice);
    }
}
@Override
public ViewHolder onCreateViewHolder(ViewGroup parent, int viewType) {
    View view = LayoutInflater.from(parent.getContext())
        .inflate(R.layout.item_row, parent, false);
    return new ViewHolder(view);
}
@Override
public void onBindViewHolder(ViewHolder holder, int position) {
    holder.txtName.setText(foodList[position]);
    holder.txtPrice.setText(foodPriceList[position]);
}
@Override
public int getItemCount() {
    return foodList.length;
}
}

```

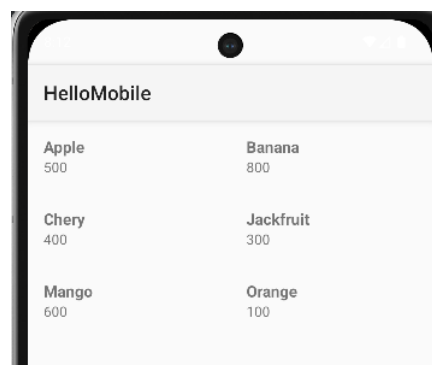
item_row.xml

```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView
        android:id="@+id/txtName"
        android:layout_width="wrap_content"
        android:textSize="16sp"
        android:textStyle="bold"
        ...../>
    <TextView
        android:id="@+id/txtPrice"
        android:textSize="14sp"
        ...../>
</LinearLayout>

```

Output :



7. SQLite Database Integration

Develop an Android application to store user information (ID, Name, Email) using SQLite. Implement insert, update, delete, and view operations and display the records in a list.

Code:

MainActivity.java

```
import android.database.Cursor;
import android.widget.*;
.....
public class MainActivity extends AppCompatActivity {
    DBHelper myDb;
    EditText editId, editName, editEmail;
    Button btnAdd, btnView, btnUpdate, btnDelete;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        myDb = new DBHelper(this);

        editId = findViewById(R.id.editId);
        editName = findViewById(R.id.editName);
        editEmail = findViewById(R.id.editEmail);
        btnAdd = findViewById(R.id.btnAdd);
        btnView = findViewById(R.id.btnView);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);
        addData();
        viewAll();
        updateData();
        deleteData();
    }

    public void addData() {
        btnAdd.setOnClickListener(v -> {
            boolean inserted = myDb.insertData(
                editName.getText().toString(),
                editEmail.getText().toString()
            );

            Toast.makeText(MainActivity.this,
                inserted ? "Data Inserted" : "Insert Failed",
                Toast.LENGTH_SHORT).show();
        });
    }

    public void viewAll() {
        btnView.setOnClickListener(v -> {
            Cursor res = myDb.getAllData();
            if (res.getCount() == 0) {
```

```

        showMessage("Error", "No Data Found");
        return;
    }

    StringBuilder buffer = new StringBuilder();
    while (res.moveToNext()) {
        buffer.append("ID: ").append(res.getString(0)).append("\n");
        buffer.append("Name: ").append(res.getString(1)).append("\n");
        buffer.append("Email: ").append(res.getString(2)).append("\n\n");
    }

    showMessage("User Data", buffer.toString());
});
}

public void updateData() {
    btnUpdate.setOnClickListener(v -> {
        boolean isUpdated = myDb.updateData(
            editId.getText().toString(),
            editName.getText().toString(),
            editEmail.getText().toString()
        );

        Toast.makeText(MainActivity.this,
            isUpdated ? "Data Updated" : "Update Failed",
            Toast.LENGTH_SHORT).show();
    });
}

public void deleteData() {
    btnDelete.setOnClickListener(v -> {
        Integer deletedRows = myDb.deleteData(editId.getText().toString());
        Toast.makeText(MainActivity.this,
            deletedRows > 0 ? "Data Deleted" : "Delete Failed",
            Toast.LENGTH_SHORT).show();
    });
}

public void showMessage(String title, String message) {
    AlertDialog.Builder builder = new AlertDialog.Builder(this);
    builder.setCancelable(true);
    builder.setTitle(title);
    builder.setMessage(message);
    builder.show();
}
}
}

```

activity_main.xml

```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"

```

```

        android:orientation="vertical"
        android:padding="16dp"
        android:fitsSystemWindows="true">

        <EditText
            android:id="@+id/editId"
            android:hint="ID"
            android:inputType="number"
            ...../>

        <EditText
            android:id="@+id/editName"
            android:hint="Name"
            ...../>

        <EditText
            android:id="@+id/editEmail"
            android:hint="Email"
            android:inputType="textEmailAddress"
            ..... />

        <Button
            android:id="@+id/btnAdd"
            android:text="Add"
            ...../>

        <Button
            android:id="@+id/btnUpdate"
            android:text="Update"
            ..... />

        <Button
            android:id="@+id/btnDelete"
            android:text="Delete"
            ...../>

        <Button
            android:id="@+id/btnView"
            android:text="View"
            ..... />
    </LinearLayout>

```

DBHelper.java

```

import android.database.Cursor;
import android.database.sqlite.*;
.....
public class DBHelper extends SQLiteOpenHelper {

    public static final String DB_NAME = "UserDB.db";
    public static final String TABLE_NAME = "users";
    public static final String COL_1 = "ID";
    public static final String COL_2 = "NAME";
    public static final String COL_3 = "EMAIL";

```

```

public DBHelper(Context context) {
    super(context, DB_NAME, null, 1);
}

@Override
public void onCreate(SQLiteDatabase db) {
    db.execSQL("CREATE TABLE users (ID INTEGER PRIMARY KEY
AUTOINCREMENT, NAME TEXT, EMAIL TEXT)");
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS users");
    onCreate(db);
}

public boolean insertData(String name, String email) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues cv = new ContentValues();
    cv.put(COL_2, name);
    cv.put(COL_3, email);
    long result = db.insert(TABLE_NAME, null, cv);
    return result != -1;
}

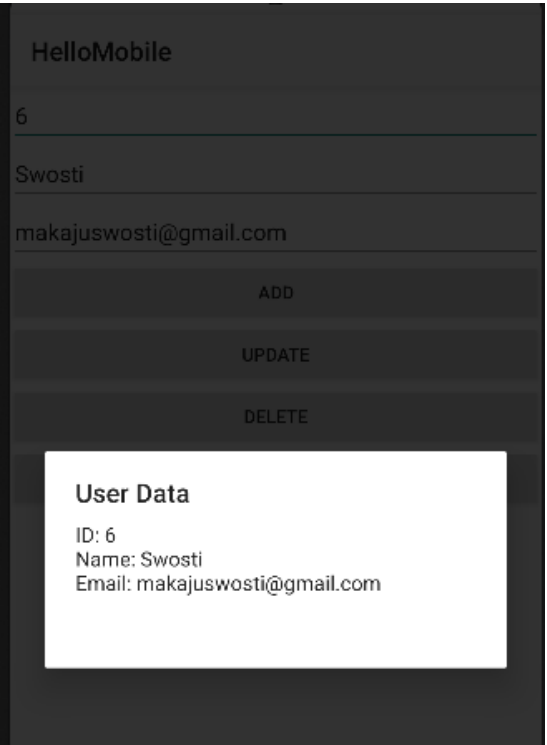
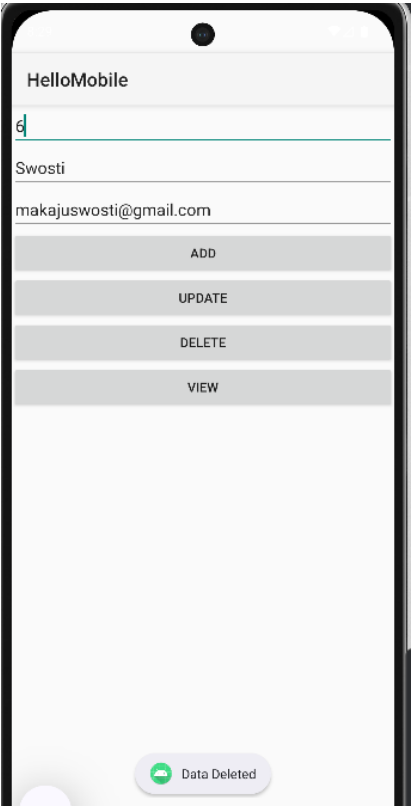
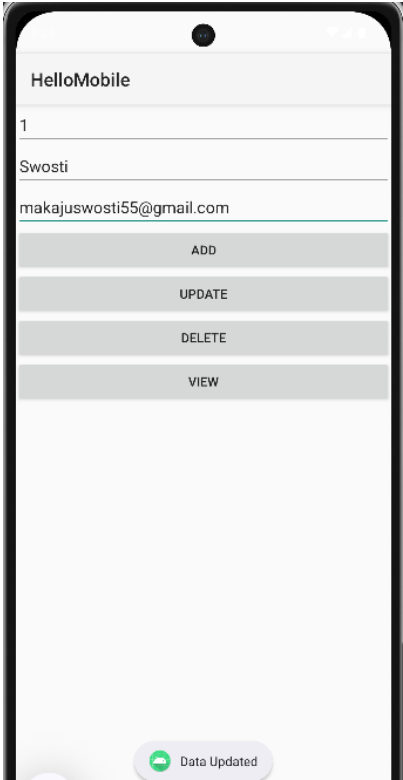
public boolean updateData(String id, String name, String email) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues cv = new ContentValues();
    cv.put(COL_2, name);
    cv.put(COL_3, email);
    db.update(TABLE_NAME, cv, "ID=?", new String[]{id});
    return true;
}

public Integer deleteData(String id) {
    SQLiteDatabase db = this.getWritableDatabase();
    return db.delete(TABLE_NAME, "ID=?", new String[]{id});
}

public Cursor getAllData() {
    SQLiteDatabase db = this.getWritableDatabase();
    return db.rawQuery("SELECT * FROM users", null);
}
}

```


Output :



8. API Integration

Create an Android application that fetches JSON data from a public REST API and displays the data in a RecyclerView.

MyApp.java

```
package com.example.hellomobile;

import android.app.Application;

import com.android.volley.RequestQueue;
import com.android.volley.toolbox.Volley;

public class MyApp extends Application {
    private static RequestQueue requestQueue;

    @Override
    public void onCreate() {
        super.onCreate();
        requestQueue = Volley.newRequestQueue(this);
    }

    public static RequestQueue getRequestQueue() {
        return requestQueue;
    }
}
```

ApiLabActivity.java

```
import androidx.recyclerview.widget.*;
import com.android.volley.*;
import org.json.*;
import java.util.*;

public class ApiLabActivity extends AppCompatActivity {
    private static final String URL = "https://jsonplaceholder.typicode.com/users";
    private UserAdapter adapter;
    private List<User> userList;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_api_lab);

        RecyclerView recyclerView = findViewById(R.id.recyclerViewApiLab);
        recyclerView.setLayoutManager(new LinearLayoutManager(this));

        userList = new ArrayList<>();
        adapter = new UserAdapter(userList);
        recyclerView.setAdapter(adapter);

        fetchUsers();
    }

    private void fetchUsers() {
        RequestQueue requestQueue = MyApp.getRequestQueue();

        JsonArrayRequest jsonArrayRequest = new JsonArrayRequest(
            Request.Method.GET,
            URL,
```

```

        null,
        response -> {
            for (int i = 0; i < response.length(); i++) {
                try {
                    JSONObject userObject = response.getJSONObject(i);
                    int id = userObject.getInt("id");
                    String name = userObject.getString("name");
                    String email = userObject.getString("email");

                    userList.add(new User(id, name, email));
                } catch (JSONException e) {
                    Log.e("API_LAB", "JSON parsing error", e);
                }
            }
            adapter.notifyDataSetChanged();
        },
        error -> {
            Log.e("API_LAB_ERROR", error.toString());
            Toast.makeText(ApiLabActivity.this, "Failed to fetch data",
                Toast.LENGTH_SHORT).show();
        }
    );

    requestQueue.add(jsonArrayRequest);
}
}

```

activity_api_lab.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fitsSystemWindows="true">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerViewApiLab"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

User.java

```

public class User {
    private int id;
    private String name;
    private String email;

    public User(int id, String name, String email) {
        this.id = id;
        this.name = name;
        this.email = email;
    }
}

```

```

    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public String getEmail() {
        return email;
    }
}

```

UserAdapter.java

```

import android.view.*;
import android.widget.*;
import androidx.recyclerview.widget.*;
import com.example.hellomobile.R;

public class UserAdapter extends RecyclerView.Adapter<UserAdapter.UserViewHolder> {

    private List<User> userList;

    public UserAdapter(List<User> userList) {
        this.userList = userList;
    }

    @NonNull
    @Override
    public UserViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
        View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.item_user, parent, false);
        return new UserViewHolder(view);
    }

    @Override
    public void onBindViewHolder(@NonNull UserViewHolder holder, int position) {
        User user = userList.get(position);
        holder.userName.setText(user.getName());
        holder.userEmail.setText(user.getEmail());
    }

    @Override
    public int getItemCount() {
        return userList.size();
    }

    static class UserViewHolder extends RecyclerView.ViewHolder {
        TextView userName, userEmail;

        public UserViewHolder(@NonNull View itemView) {
            super(itemView);
            userName = itemView.findViewById(R.id.userName);
            userEmail = itemView.findViewById(R.id.userEmail);
        }
    }
}

```

Item_user.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="8dp"
    app:cardCornerRadius="8dp"
    app:cardElevation="4dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:id="@+id/userName"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textSize="18sp"
            android:textStyle="bold"
            android:textColor="@android:color/black" />

        <TextView
            android:id="@+id/userEmail"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="4dp"
            android:textSize="14sp"
            android:textColor="@android:color/darker_gray" />

    </LinearLayout>
</androidx.cardview.widget.CardView>
```

HelloMobile

Leanne Graham

Sincere@april.biz

Ervin Howell

Shanna@melissa.tv

Clementine Bauch

Nathan@yesenia.net

Patricia Lebsack

Julianne.OConner@kory.org

Chelsey Dietrich

Lucio_Hettinger@annie.ca

Mrs. Dennis Schulist

Karley_Dach@jasper.info

Kurtis Weissnat

Telly.Hoeger@billy.biz